



Approved Crystals for NanoBeacon SoC IN100

Control number:
Revision: V1.8



Contents

1. Summary.....	4
2. Approved 26MHz Crystal.....	6
3. Vendor Contact information	9
4. Revision History	11
5. Disclaimer	11
6. Appendix A	12
7. Appendix B.....	14
8. Appendix C.....	16
9. Appendix D.....	18
10. Appendix E.....	20
11. Appendix F	21
12. Appendix G.....	22

 List of Figure

Figure 1: NanoBeacon Config Tool for Crystal Setting	4
Figure 2: IN100 QFN18 reference design	6

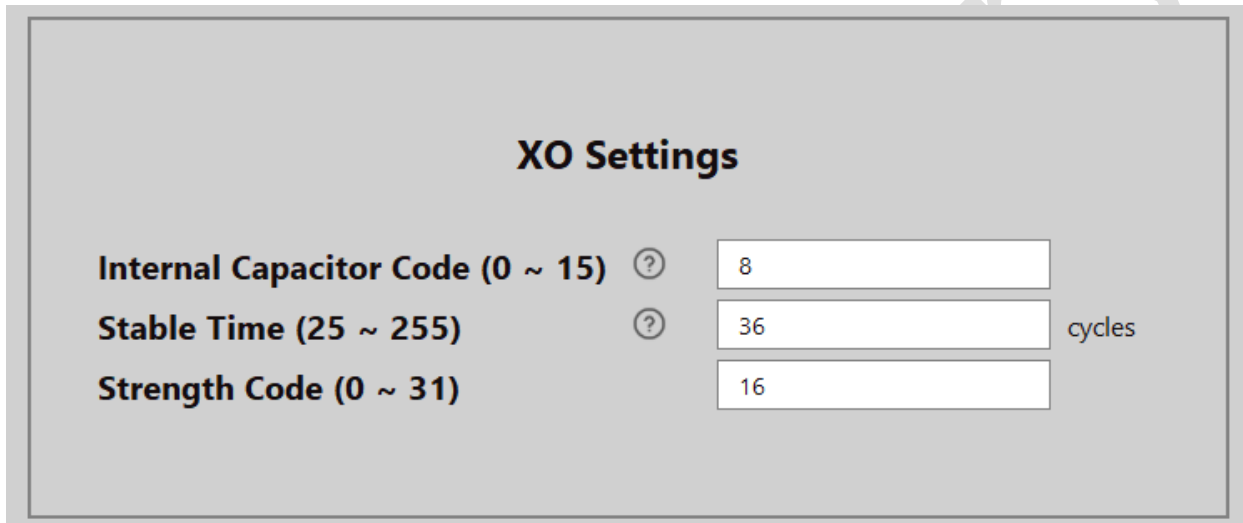
 List of Tables

Table 1 : IN100 Product list	5
Table 2: Approved 26MHz crystals.	7
Table 3: Approved 26MHz Crystal Setting on IN100DK	8

1. Summary

This document describes some strategies for developers to properly select and configure 26MHz crystals when developing InPlay NanoBeacon SoC IN100-based systems, explains which crystal parameters affect the system, and provides a list of several verified recommended reference crystal manufacturers and their part numbers.

Three parameters are critical to the stable operation of the IN100 based system. Figure 1 shows the IN100 default settings for the 26MHz crystal.



XO Settings	
Internal Capacitor Code (0 ~ 15) ?	8
Stable Time (25 ~ 255) ?	36 cycles
Strength Code (0 ~ 31)	16

Figure 1: NanoBeacon Config Tool for Crystal Setting

Internal Capacitor Code: It is used for frequency bias calibration. Adjusting this value is the process of adjusting the frequency of the crystal oscillator to match an exact reference frequency. The accuracy of the crystal oscillator depends on the accuracy of the crystal resonator itself and the temperature stability of the oscillator circuit. Note: Adjusting the capacitance value of the on-chip crystal oscillator will affect the start-up time and drive strength.

Stable Time: Crystal stabilization time is the time required for a crystal oscillator to reach its stable operating frequency after power is applied. A long stabilization time will result in longer system start-up time and will consume more power, while a short stabilization time will reduce the reliability of the system and causes the chip to fail to start. The stabilization time of a crystal oscillator is affected by various factors, such as the drive strength (the amount of current applied to the crystal), the type and size of the crystal, and the oscillator circuit design. Optimizing the crystal start-up time requires a careful balance between drive strength, crystal specifications, and oscillator circuit design.

Strength Code: The drive strength is the amount of current applied to the crystal to drive it to oscillate at its resonant frequency. A higher drive strength reduces the crystal's stabilization time, but it also increases power consumption and may damage the crystal. The optimal drive strength of a crystal oscillator depends on various factors, such as the type and size of the crystal, and the operating conditions.

This document applies to the following products:

Table 1 : IN100 Product list

Applicable products	Product status
IN100-D1-R-RC1I	Mass production
IN100-D1-R-YC1I	Mass production
IN100-Q1-R-RC1I	Mass production
IN100-Q1-R-YC1I	Mass production
IN100-Q1-R-YC1F	Mass production
IN100-W10-R-SC1I	Mass production

2. Approved 26MHz Crystal

For all approved crystals, no external load caps are required. Frequency calibration only requires trimming the internal cap code of IN100. **Do not install C4 and C5** when using approved crystals.

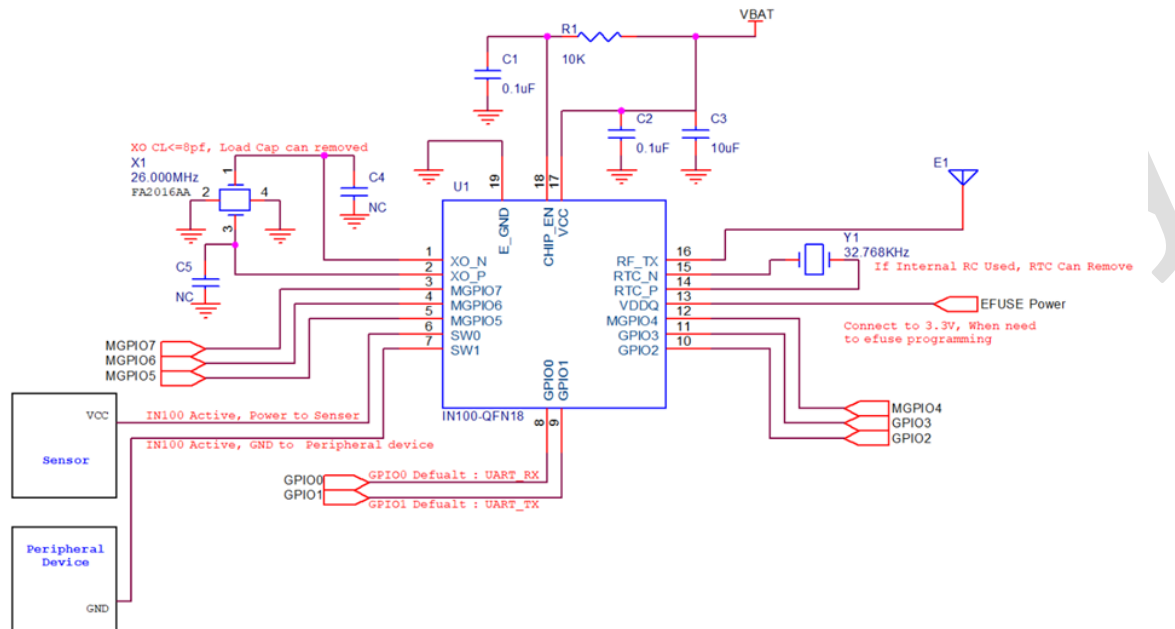


Figure 2: IN100 QFN18 reference design

Table 2 lists the approved crystals for the IN100.

Table 2: Approved 26MHz crystals.

N o	Part number	Vender	Package Size	CL	C0	ESR	Temp Range
1	CN4026M00006BE840000	JWT	2.0 x 1.6 x 0.5 mm	6p F	≤2p F	≤40oh m	-40°C ~ +125°C
2	CN4026M000067A840000	JWT	2.0 x 1.6 x 0.5 mm	6p F	≤2p F	≤40oh m	-40°C ~ +85°C
3	XRCGE26M000FXA2AR0	Murata	2.0 x 1.6 x 0.65 mm	8p F	≤3p F	≤60oh m	-40°C ~ +125°C
4	XRCGB26M000F1SBLR0	Murata	2.0 x 1.6 x 0.7 mm	7p F	≤3p F	≤40oh m	-40°C ~ +105°C
5	ABM12L-0000-T3	Abracon	1.6 x 1.2 x 0.33 mm	8p F	≤2p F	≤80oh m	-40°C ~ +85°C
6	ABM11W-0018-T3	Abracon	2.0 x 1.6 x 0.5 mm	8p F	≤2p F	≤80oh m	-40°C ~ +85°C
7	LFXTAL081614	IQD	1.6 x 1.2 x 0.35 mm	8p F	≤7pF	≤70oh m	-40°C ~ +85°C
8	LFXTAL082126	IQD	2.0 x 1.6 x 0.5 mm	8p F	≤7pF	≤25oh m	-40°C ~ +85°C
9	TXC 8Y26070026	TXC	2.0 x 1.6 x 0.5 mm	8pf	3.5p f	60ohm	-40°C ~ +85°C
10	XTL501150-B89-019	Siward	2.0 x 1.6 x 0.5 mm	8p F	≤3p F	≤60oh m	-30°C ~ +85°C
11	NX2016SA EXS00A- CS16966	NDK	2.0 x 1.6 x .45 mm	6pf	≤3p F	≤60oh m	-40°C ~ +85°C

For more test results, please refer to appendices A through E.

Note:

- 1* All test setups are at room temperature using IN100. 3.3V supply voltage is used.
- 2* Default settings are used for XO setting of stabilization time and drive intensity code.
- 3* Frequency is calibrated by the IN100's internal cap code

Table **3** lists the approved crystals' XO settings for IN100 config tool GUI. Users will need to adjust the cap code according to the actual hardware situation.

Performance below was measured at the following conditions for the IN100 DK board: $T_a = 25^{\circ}\text{C}$, $V_{CC} = 3.3\text{V}$, unless otherwise noted.

Preliminary

Table 3: Approved 26MHz Crystal Setting on IN100DK

No	Part number	Vender	IN100 Capacitor Code	Stable Time	Strength Code
1	CN4026M00006BE840000	JWT	5	36	16
2	CN4026M000067A840000	JWT	5	36	16
3	XRCGE26M000FXA2AR0	Murata	10	36	25
4	XRCGB26M000F1SBLR0	Murata	7	36	16
5	ABM12L-0000-T3	Abracon	11	45	16
6	ABM11W-0018-T3	Abracon	11	45	16
7	LFXTAL081614	IQD	11	36	20
8	LFXTAL082126	IQD	10	36	18
9	TXC 8Y26070026	TXC	10	36	18
10	XTL501150-B89-019	Siward	10	36	16
11	NX2016SA EXS00A-CS16966	NDK	4	36	16

3. Vendor Contact information

- **JWT**
Company: HEFEI JINGWEITE ELECTRONICS CO.,LTD.
Website: <http://www.hfjwt.cn>
Address: No.2569, Yungu Road, Economy & Technology Area,Hefei ,Anhui, China
Telephone: +86 0551-63350130,63350092
Contact: Aaron Guo
Mobile: +86-156 5656 9669
Email: gxc@hfjwt.cn
- **Murata**
Company: Murata
Website: murata.com
Telephone: +1 4082506477
Mobile: +1 4082506477
Contact: Dustin DeFrank
Email: dustin.defrank@murata.com
- **Abracon**
Company: ABRACON LLC
Website: www.abracon.com
Address: 5101 Hidden Creek Lane Spicewood, TX 78669
Telephone: +1(512)371-6159
Fax: (512) 351-8858
- **IQD**
Company: IQD Frequency Products Ltd
Website: www.iqdfrequencyproducts.com
Address: Station Rd, Crewkerne TA18 8AR, United Kingdom
Telephone: +44 1460-270200
Email: Sales@iqdfrequencyproducts.com
- **TXC**
Company: TXC Corporation
Website: www.txccorp.com
Address: No.4, Kung Yeh 6th Rd., Ping Cheng District 324403,Tao Yuan City, Taiwan, R.O.C.
Telephone: +886-3-4698121
- **Siward**
Company: Siward
Website: www.sward.com
Address: No. 1-1, Ln.111, Sec.3, Zhongshan Rd., Tanzi Dist., Taichung City 427018, Taiwan, R.O.C
Telephone: +886-4-2534-7909
Email: sales@siward.com

- **NDK**

Company: NIHON DEMPA KOGYO CO (NDK)

Website: www.NDK.com

Address: Merkmal Keio Sasazuka Bldg., 1-47-1, Sasazuka, Shibuya-ku, Tokyo, 151-8569, Japan

Telephone: +81-3-5453-6711

Email: jason.glavy@ndk.com

4. Revision History

Revision	Description	Update date	Owner
V1.0	Preliminary Version	9/1/2023	Liang X.
V1.1	Preliminary Version	9/18/2023	Aidan M.
V1.2	Updated to include latest approved xtals	10/20/2023	Aidan M.
V1.3	Updated for latest approved xtals	4/20/2024	Aidan M.
V1.4	Updated list with TXC parts and fixed formatting issues	11/20/2024	Aidan M.
V1.4	Update table voltage number to 3.3 V	9/11/2024	Aidan M.
V1.5	Updated TXC part numbers	1/10/2025	Aidan M.
V1.6	Updated Siward part numbers	3/10/2026	Aidan M.
V1.7	Updated NDK part numbers.	3/10/2026	Aidan M.
V1.8	Updated IQD part numbers	6/5/2026	Aidan M.

5. Disclaimer

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6. Appendix A

JWT CN4026M00006T2998001 test results as follows:

频率测试 (Measure the FREQ. of crystal on PCB board) :

初始测试 (Original)

1、在PCB上测试输出频率 (Measure the output frequency on PCB board) Fout1:

输出频率 (Fout1) = 25.999908 MHz, -3.54 ppm

2、晶振性能测试 (室温) Crystal characteristic test (at room ter 26°C

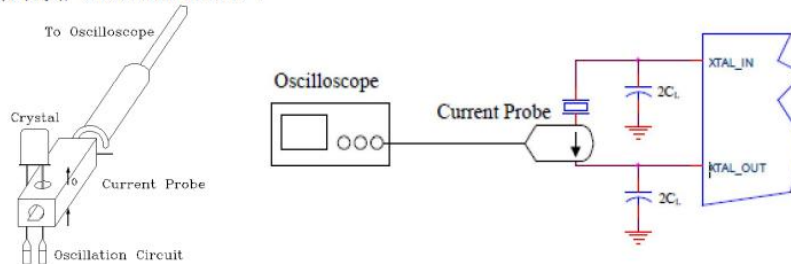
Specification(CL1) = 6 pF

FL1 (MHz)	FL1 (ppm)	Rs (ohm)	C0 (pF)	C1 (fF)	Ts (ppm/pF)
26.000034	1.29	18.53	0.65	1.67	18.9

3、 $\Delta F = |F_{out1} - FL1| = \underline{-4.83}$ ppm

驱动功率测试 (Measure the drive level of crystal on PCB board) :

1、测试线路 (Measured Circuit) :



2、测试方法 (Measured Method) :

使用电流探头进行Vrms测量(Tek CT-1, 50Ω)

Use AC current probe to conduct Vrms measuring (Tek CT-1, 50 ohms terminated)

Irms = Vrms/5(mA)

3、计算方法 (Calculation Method) :

$$I_{rms} = \underline{0.63} \text{ mA}$$

$$R_L = R_s (1 + C_0 / C_L)^2$$

$$= \underline{18.53} * (1 + (\underline{0.65} / \underline{6}))^2 = \underline{22.76} \text{ ohms}$$

$$\text{Drive Level} = I_{rms}^2 * R_L$$

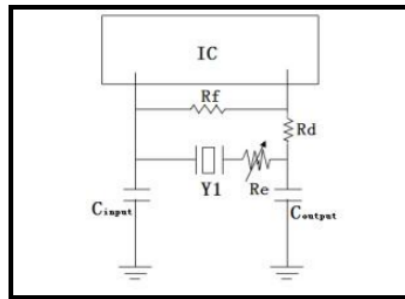
$$= \underline{0.63}^2 * \underline{22.76} = \underline{9.09} \text{ } \mu\text{W} \quad \text{Data sheet Max Level } \underline{200} \text{ } \mu\text{W}$$

Drive Level < Data sheet Max Level Good

Drive Level ≥ Data sheet Max Level Not matching

振荡电路负阻测试 (Measure the negative resistance (- R) value of chip set) :

1、测试线路 (Measured Circuit) :



2、计算方法 (Calculation Method) :

$$| -R | = R_e + R_L = \frac{270}{292.76} + \frac{22.76}{40} = \frac{292.76}{7.32} \text{ ohms}$$

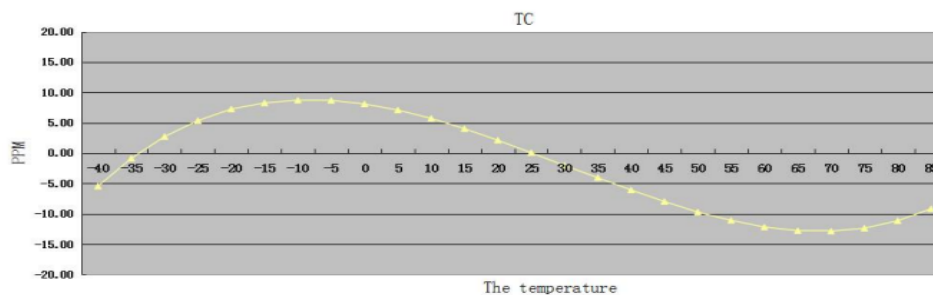
n:
Poor -R value: $n < 3$
Normal -R value: $3 < n < 5$
Good -R value: $n > 5$

RL: 负载电阻 (Load Resistance)
R: 等效电阻 (ESR)
Re: 可调电阻 (Variable Resistor)

Measured oscillation characteristics

Parameter	Measured Typical Results
Work the amplitude	558 [mv]
Maximum operating amplitude	520 [mv]
Minimum operating amplitude	620 [mv]
Drive level	6.32 [uW]
The actual load capacitance	6.0 [pF]
Temperature frequency drift	See chart below
Crystal start time	0.408 [ms]

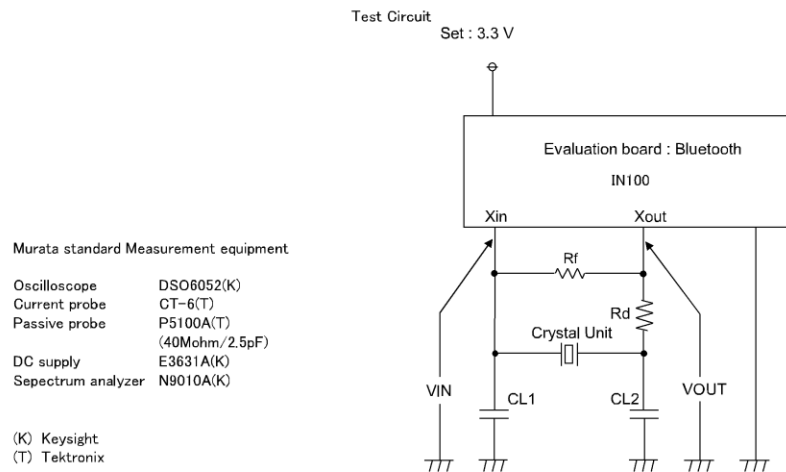
Measured frequency drift by temperature



7. Appendix B

Murata XRCGB26M000F1SBLR0 test results as follows:

Item		Condition
IC name		IC名
Parts Number of Crystal Unit		村田型号
Circuit Parameter		
External load capacitance	负载电容	CL1
	负载电容	CL2
	反馈电阻	Rf
	阻尼电阻	Rd
Supply Voltage Range		电源电压范围
Temp. Range		温度范围



Circuit Characteristics 特性	Value 测定值		Remarks 备注
Center Frequency and Difference 起振回路上起振频率与偏差量 (*1) (Typical sample at Set=3.3V,+25deg.C)	25.999905	[MHz]	Oscillating frequency and its shift against nominal frequency 在起振回路上的频率以及相对于公称频率之间的偏差量
	-4	[ppm]	
Load Capacitance on your PCB 负载容量值 (Typical sample at Set=3.3V,+25deg.C)	7.3	[pF]	This value shows load capacitance the evaluated circuit has 在起振回路上等价于连接在谐振器两端的容量
Negative Resistance and Oscillation margin 负性电阻/起振余裕度 (at Set=3.3V,+25deg.C)	R	317 [Ω]	The details is explained in page 2 详细内容参见下页说明
	Ratio	5.3 [Times]	
Drive Level 激励功率 (Typical sample at Set=3.3V,+25deg.C)	11	[uW]	Drive power of crystal under circuit condition 起振回路在工作状态下谐振器消耗的功率
Oscillating Voltage 起振电压 (Typical sample at Set=3.3V,+25deg.C)	VINp-p	0.5 [V]	Swing level at input side 输入端起振振幅 (VIN_H - VIN_L)
	VOUTp-p	0.5 [V]	Swing level at output side 输出端起振振幅 (VOUT_H - VOUT_L)
Oscillation Start up Time 启动时间 (*2) (Typical sample at Set=3.3V,+25deg.C)	0.54	[ms]	Time to reach 90% of the oscillation level under steady state 达到稳定状态振幅的90%所需要时间

*1 Frequency difference means the oscillating frequency difference between your PCB and Murata's frequency sorting circuit.

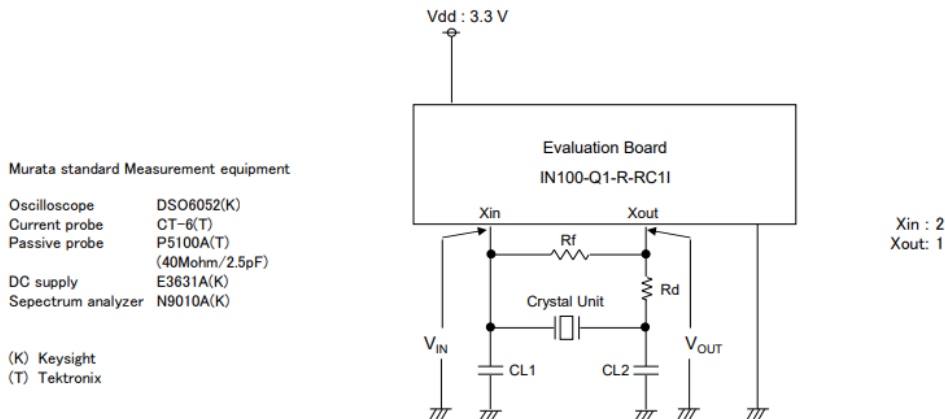
频率偏差指在贵公司基板上的测定频率与本公司标准回路上测定频率间的偏差。

*2 The measurement results is affected by the rise-up characteristics of supplied voltage on your PCB.

测定结果受安装基板上电源启动方式的影响。

Murata XRCGE26M000FXA2AR0 test results as follows:

IC name	IC名	IN100-Q1-R-RC1I
Parts Number	品番	XRCGE26M000FXA2AR0
IC's setting value	Internal Capacitance Code	10
	Stable Time	36
	Strenght Code	25
CL1 [pF]	負荷容量	Open
CL2 [pF]	負荷容量	Open
Rf [ohm]	帰還抵抗	No mount
Rd [ohm]	制限抵抗	Short
Supply Voltage Range[V]	電源電圧範囲	3.3 V
Temp. Range [deg.C]	温度範囲	-40 to +125deg.C



Characteristics 特性	Value 測定値	Criterion 基準	Notes 備考
Oscillating Voltage 発振電圧 (Vdd=3.3V,+25deg.C) (Vcel=1.2V) Vcel is driving voltage of oscillation circuit Vcelは発振回路の電圧を表します	VIH	0.8 [V]	≤ 1.5V Input high voltage 発振入力電圧High側レベル
	VIL	0.2 [V]	≥ -0.3V Input low voltage 発振入力電圧Low側レベル
	VOH	0.8 [V]	≤ 1.5V Output high voltage 発振出力電圧High側レベル
	VOL	0.1 [V]	≥ -0.3V Output low voltage 発振出力電圧Low側レベル
	VIp-p	0.5 [V]	Swing level at input side 発振入力振幅 (VIH - VIL)
	VOp-p	0.6 [V]	Swing level at output side 発振出力振幅 (VOH - VOL)
Oscillation Start up Time 発振立ち上がり時間 (*2) (Typical sample at Vdd=3.3V,+25deg.C)	0.72 [ms]		Time to reach 90% of the oscillation level under steady state 定常状態の発振振幅の90%に達するまでの時間
Center Frequency Difference 発振回路における発振周波数ずれ量 (*3) (Typical sample at Vdd=3.3V,+25deg.C) (参考値)	-2 [ppm]		Oscillating frequency shift against nominal frequency 振動子の公称周波数と発振回路における発振周波数のずれ量
Negative Resistance Analysis 発振余裕度 (at Vdd=3.3V,+25deg.C)	Rs_max [Ω] 270	R [Ω] 318 Ratio [Times] 5.3	The details is explained in page 2 詳細につきましては、次頁をご参照下さい。
Load Capacitance on your PCB 負荷容量値 (Typical sample at Vdd=3.3V,+25deg.C)	8.2 [pF]		This value shows load capacitance the evaluated circuit has. 発振回路において振動子の両端に仮想的に接続される容量
Drive Level ドライブレベル (Typical sample at Vdd=3.3V,+25deg.C)	20 [uW]		Drive power of crystal under circuit condition 発振回路が動作している状態において振動子で消費される電力

*1 R1: Equivalent series resistance. "R1 limit sample" means R1 resistance of applied sample is equal as its R1 spec.
R1は振動子の等価直列抵抗を意味します。R1規格限界品とは、適用する振動子の等価直列抵抗値が、その規格値まで低下した状態のサンプルとなります。

*2 The measurement results is affected by the rise-up characteristics of supplied voltage on your PCB.

測定結果は実装基板の電源立ち上がり特性の影響を受けます。

*3 Frequency difference means the oscillating frequency difference between your PCB and Murata's frequency sorting circuit.

Please refer to appendix 1.

貴社基板と当社検査回路では、同一振動子を使用した場合でも発振周波数の差が生じます。これを発振周波数相間と呼びます。添付資料1をご参照下さい。

8. Appendix C

Abracon ABM12L-0000-T3 test results as follows:

Series-Parallel resonance & motional parameters @ (+25°C ± 3°C)

Crystal Parameters			
Nominal Frequency	Fnom	26,000,000	Hz
Parallel Resonance	FL	25,999,937	Hz
Motional resistance	Rm	24.64	Ω
Motional inductance	Lm	34.17	mH
Motional capacitance	Cm	1.10	fF
Static capacitance	C0	0.91	pF

In-Circuit Frequency

Basic Oscillator Parameters			
Nominal Frequency	Fnom	26,000,000	Hz
Crystal Parallel Resonance	FL	25,999,937	Hz
Operating Frequency	F	25,999,926	Hz
Internal Capacitor Code	(0~15)	11	

Negative Resistance and Safety Factor

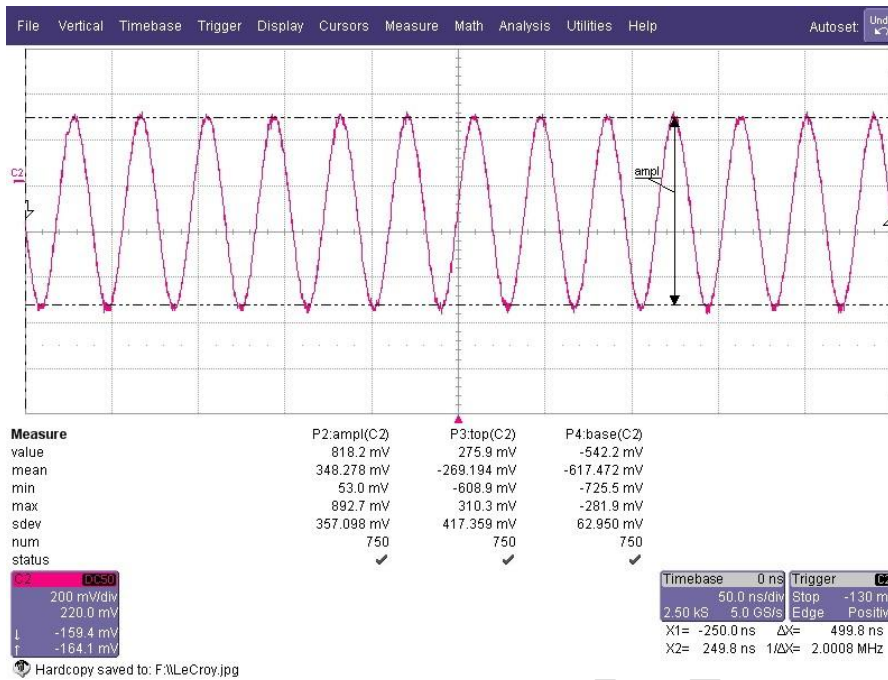
Basic Oscillator Parameters			
Crystal Motional resistance	Rm	24.7	Ω
Maximum Crystal Resistance	Max ESR	80	Ω
Added Series Resistance	Rs	400	Ω
Negative Resistance	-R	>424	Ω
Safety Factor (Ratio)	SF	>5	

Oscillator loop continues to operate beyond an added series resistance of 400 ohms.

Drive Level

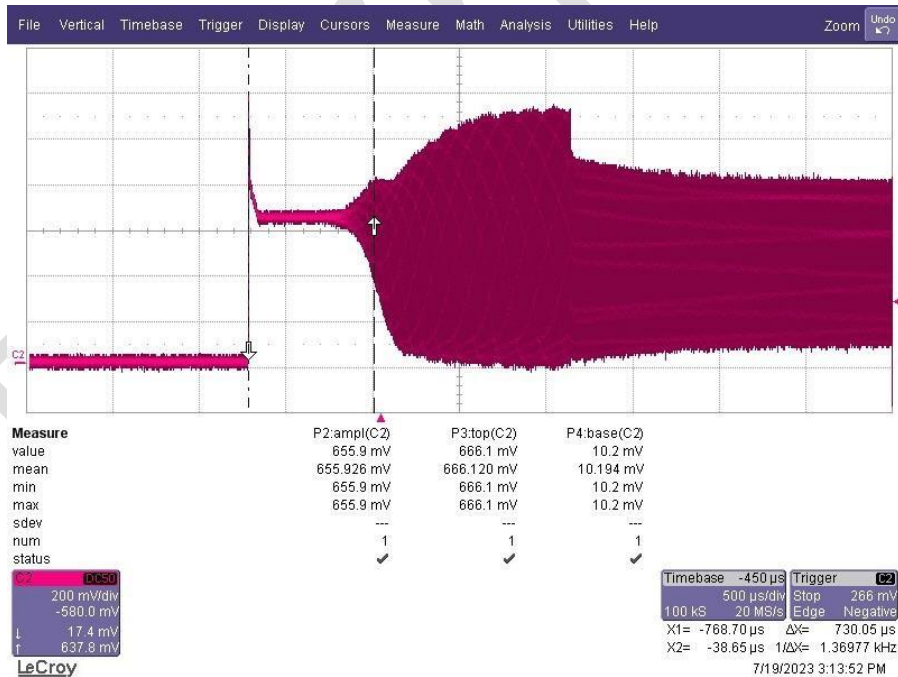
Basic Oscillator Parameters			
Crystal Effective Series Resistance	ESR	26.3	Ω
Through-Crystal Current	I	0.7	mA
Drive Level	DL	11.5	μW

Oscillation Waveform



Passed Waveform Testing: Amplitude > 550 mV

Oscillation Start-up Waveform



Passed Startup Testing: Amplitude > 400mV in 730μs (Stable Time = 45)

9. Appendix D

IQD LFX TAL081614 test results as follows:

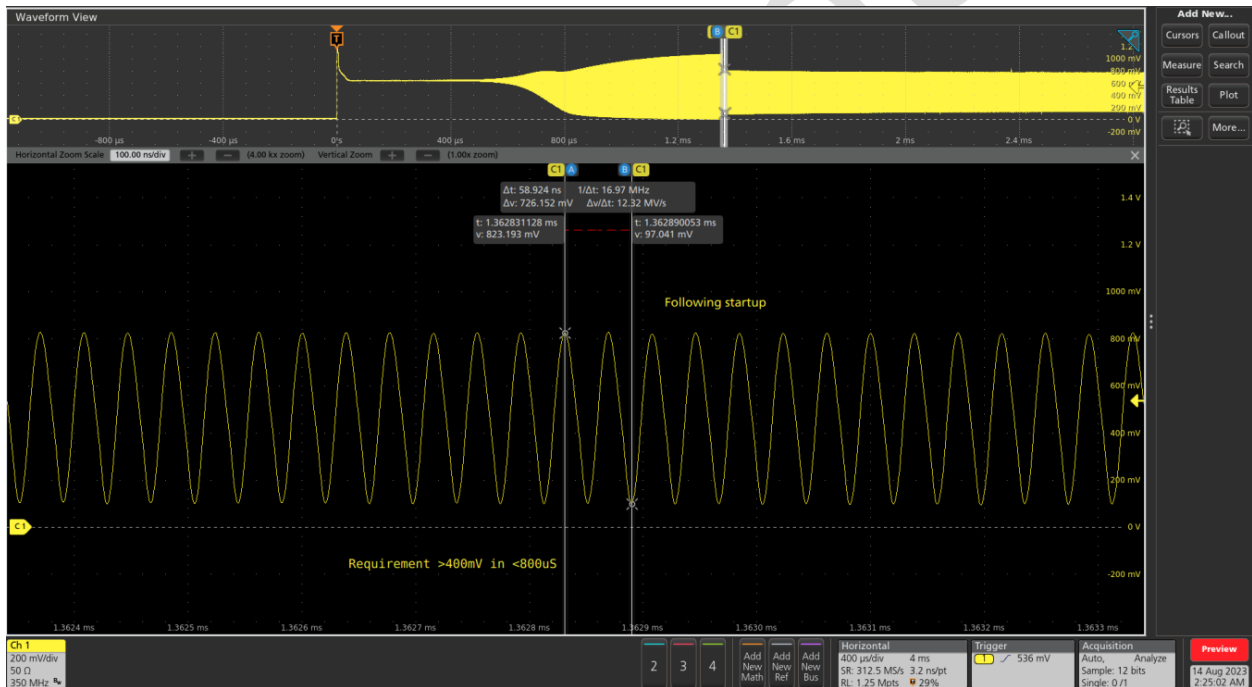
PPM Results

Part Tested	PCB Identifier	IC Fitted	Xtal Manufacturer	Fundamental frequency Hz	Measured Frequency Hz On PCB	Δ Hz	Calculated ppm deviation in circuit ppm
LFX TAL081614	PCB #1-81614	IN100	IQD	26000000.00	25999720.7636220	-279.2363780	-10.74
LFX TAL081614	PCB #2-81614	IN100	IQD	26000000.00	25999745.9191951	-254.0808049	-9.77

$$ppm = \frac{\text{measured Hz} - \text{fundamental Hz}}{\text{fundamental Hz}} * 1000000$$

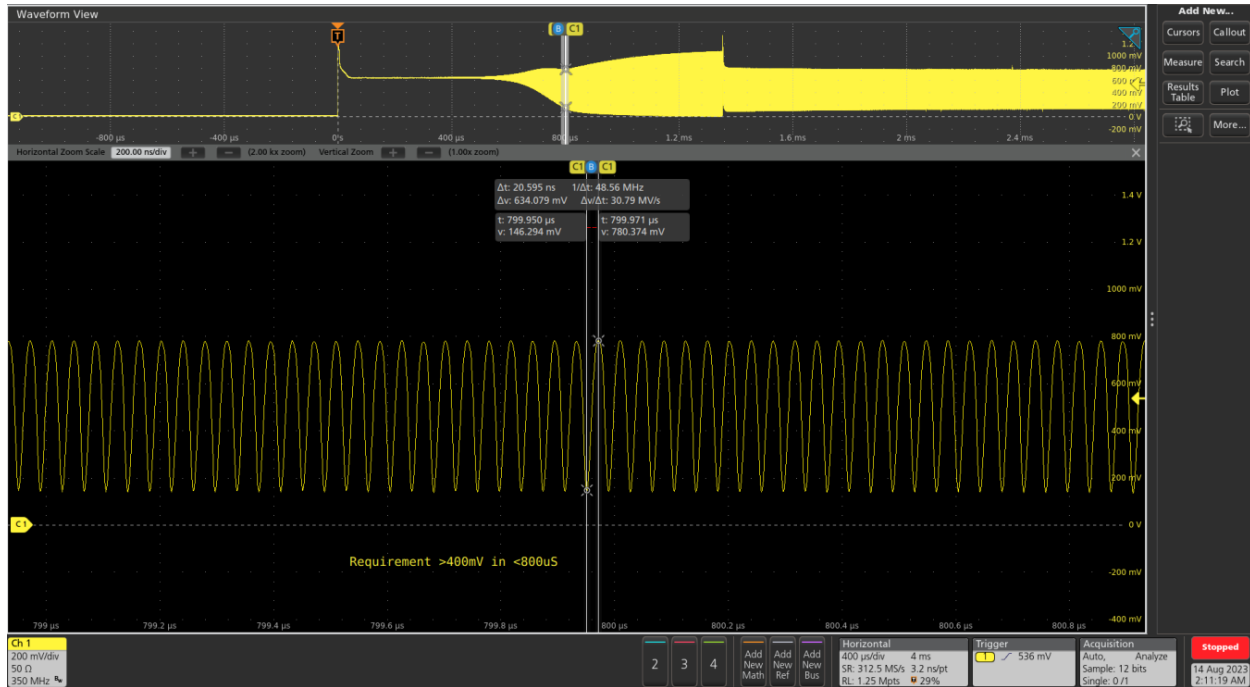
Passed BLE PPM requirements

Oscillation Waveform



Passed oscillation waveform test: Greater than 550 mV amplitude

Oscillation Startup Waveform



Passed startup requirements: 400 mV or greater amplitude in 800 us or less.

Negative Resistance Ratio and ESR:

Resistance at failure to oscillate Ohms	Calculated Neg Ratio Based upon measured ESR	Neg R Ratio based upon Max ESR in datasheet	Max ESR from datasheet
301.20	4.30	1.51	200

Figure 8 - Negative resistance ratio with ESR measured at 70Ω

$$\text{Negative resistance ratio} = \frac{\text{Resistance at point of poor startup or failure to start}}{\text{ESR from datasheet}}$$

It can be seen that the negative resistance ratio in this part is below 5 when we consider the maximum advertised ESR of 200Ω, however IQD historical measured data shows this part does not exceed 70Ω which gives a ratio of 4.3 times.

Passed negative resistance ratio: negative resistance ratio between 4 and 5. *Note that measured ESR by IQD was 70 Ohms, while datasheet number is an overestimation according to IQD engineer.

10. Appendix E

TXC test results are as follows:

Characteristic and PPM Results:

1. Crystal Unit room temperature(25°C) characteristic test by S&A 250B Network analyzer

Vendor	FL(ppm)	Rr(Ω)	C0(pF)	C1(fF)	L(mH)	TS(ppm/pF)
TXC	3.19	15.60	0.58	1.67	22.45	11.34

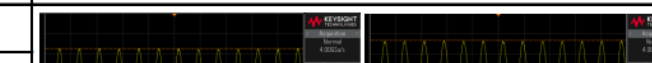
PPM and other characteristics are found to be within BLE and IN100 requirements.

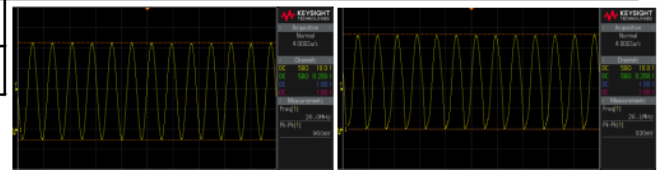
Board Tests: Amplitude, ESR, and Negative Resistance Ratio Tests:

2. Crystal on board test.

C16,C15=DNI

Internal Capacitor Code(0~15) : 10

Vendor	Output Frequency (Hz)	Output Frequency (ppm)	$\triangle F$ (ppm)	DL (uW)	Irms (mA)	Starting current DL (uW)	Starting current Irms (mA)	VP-P(V)
TXC	25999980	-0.77	-3.96	12.77	0.85	9.68	0.74	0.96/0.93
Ratio : 6.47 Ratio = $ -R (\Omega)/Rr(\Omega)$ spec				$ -R (\Omega) : 388$		$ -R (\Omega)=R_s_max + ESR(\Omega)$		
Rr(Ω) spec : 60				Rs_max : 370 ESR(Ω) : 17.68				
ESR(Ω) : 17.68 ESR(Ω)= Rr(Ω) *(1+C0/CL) ²								
Rr(Ω) : 15.60 C0(pF) : 0.58 CL : 9.01								

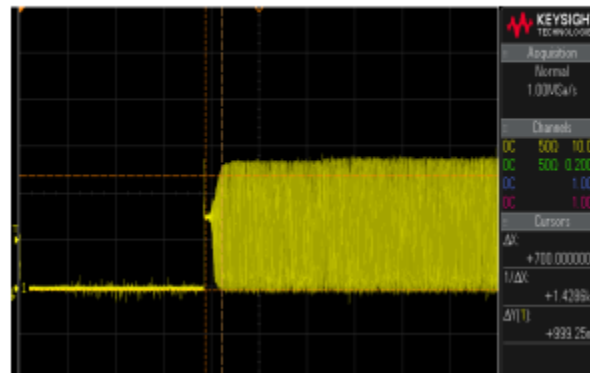


$$DL = I_{rms}^2 * ESR$$

ESR is within requirements and the resulting negative resistance ratio exceeds the required value of 4 to 5. Amplitude is shown to be above 550 mV at all relevant points in the xtals oscillation time.

Crystal Startup Results:

4.) Start time :



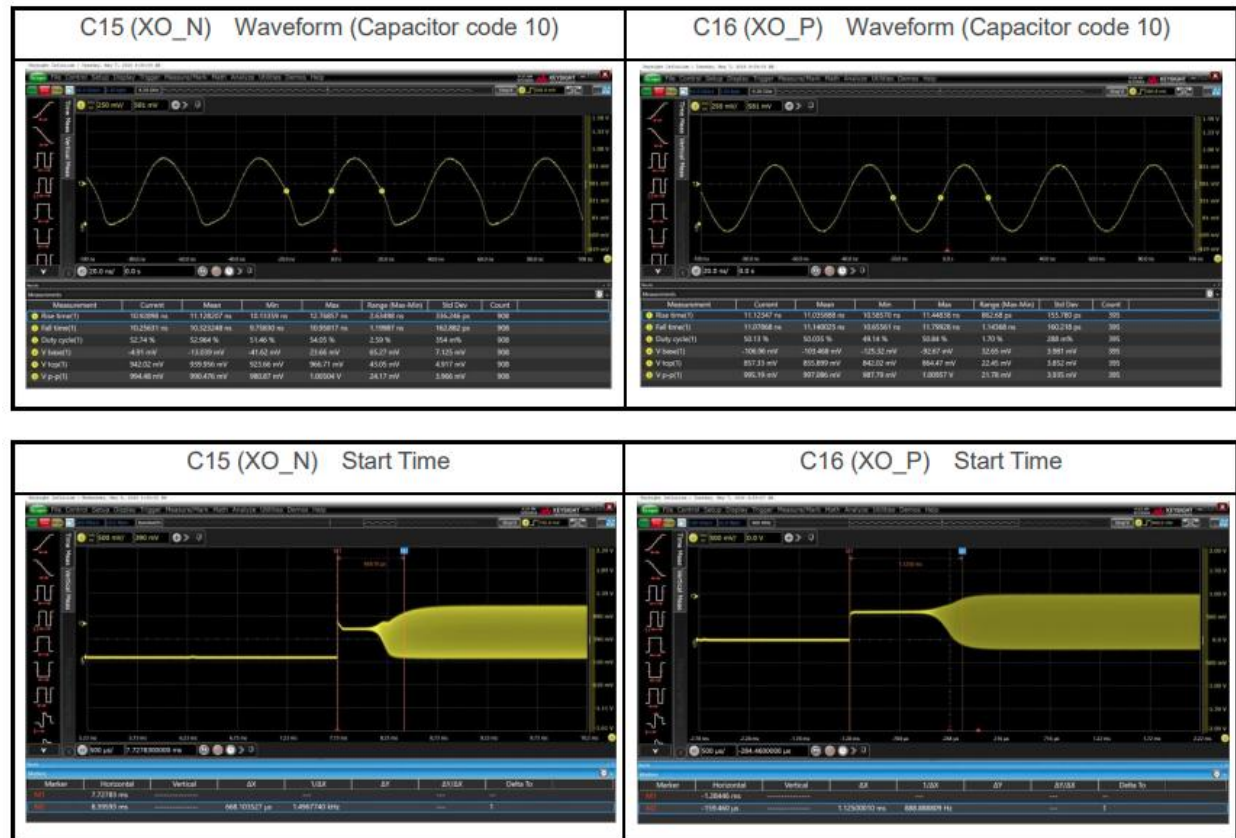
Startup time is much faster than 800 us requirement.

11. Appendix F

The Siward results are as follows:

Circuit	Circuit BOM list			CL [pF]	-R [Ω]	-R/R1 _{max} ≥5	DL [μW]	Judgment
	C15 [pF]	C16 [pF]	Internal Capacitor Code [0~15]					
Original	DNI	DNI	10	8.0	-518	8.6	15	○
Crystal Specification				8.0pF		R1: 60Ωmax		

Internal cap code is 10 with a measured negative resistance ratio of 8.6



Waveforms for amplitude and startup are within InPlay specifications. Frequency deviation is -0.2 ppm.

12. Appendix G

NDK results are as follows:

EXC10B-85973A-2/5

3. RESULTS

- 1) Recommended Negative Resistance Target: $-R = 1000 \Omega$ minimum. (NDK target)
- 2) The circuit has the following characteristics:
 $\langle \text{EXS00A-CS16966, CL}=6\text{pF, measured Xtal ESR}=30 \text{ ohm, Max. ESR of spec.}=60 \text{ ohm} \rangle$

Circuit Resister setting	Internal capacitance [pF]	Cin [pF]	Cout [pF]	Negative Resistance -R [Ω]	Negative Resistance ratio from max. ESR	Frequency Deviation dF/F [$\times 10^{-6}$]	Drive Level DL [uW]	Start up time [us]	Output waveform amplitude (mVp-p)
					>4	< +/-50	<200	<800	>550
0	0.5	NM	NM	1490	24.8	+57	12	530	1150
1	1.0	NM	NM	1110	18.5	+39	13	460	1110
2	1.5	NM	NM	850	14.2	+25	15	430	1060
3	2.0	NM	NM	660	11.1	+13	16	450	1010
4	2.5	NM	NM	510	8.5	+4	16	460	950
5	3.0	NM	NM	450	7.5	-5	16	500	910
6	3.5	NM	NM	340	5.8	-13	15	500	840
7	4.0	NM	NM	300	4.9	-19	15	460	810
8	4.5	NM	NM	270	4.4	-26	14	430	740
9	5.0	NM	NM	220	3.6	-32	13	460	660
10	5.5	NM	NM	190	3.2	-37	12	440	590
11	6.0	NM	NM	160	2.6	-42	11	460	500
12	6.5	NM	NM	150	2.5	-45	9	490	450
13	7.0	NM	NM	140	2.3	-49	7	440	400
14	7.5	NM	NM	110	1.9	-52	6	500	380
15	8.0	NM	NM	100	1.7	-55	5	440	360

* DL:200uW Max.

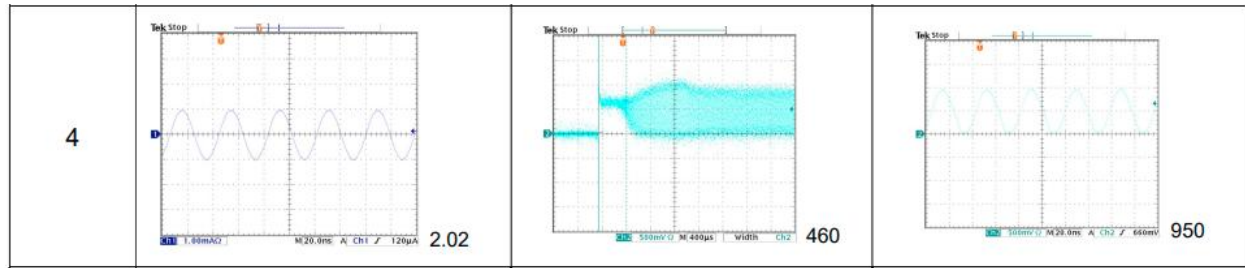
- Formula
 - Negative resistance ratio = Negative resistance / max ESR
 - Max ESR= 60 (ohm)
 - Drive level DL

Formula

$$I_{RMS} = \frac{I_{P-P}}{2\sqrt{2}}$$

$$P = I_{RMS}^2 R_{Load} [\text{uW}] \leftarrow \text{Drive level}$$

The table above shows the best result with a capacitor code of 4 with a PPM of +4, a negative resistance ratio of >8, and both startup and waveform amplitude within InPlay specs.



The results above show the oscilloscope results reported for cap code 4 matching the table.